

Homework #5

Due 2:50 pm, Thursday, Nov. 10th, 2022

1. Consider a simple 4x16, 8-bit image:
 22 22 21 95 93 94 166 167 166 167 253 254 254 255 253 55
 20 22 21 22 95 95 93 166 167 166 254 253 255 255 254 54
 21 21 20 22 95 95 95 165 166 166 255 254 254 254 253 54
 21 21 22 95 95 94 166 167 165 166 254 255 253 253 255 55
 (a) Compute the entropy of the image. (5 pts)
 (b) Compress the image using Huffman coding and report the compression ratio using Huffman coding. (10pts)
 (c) Use another data compression algorithm shown in the lecture to obtain a higher compression ratio. (10 pts)

Solution:

- (a) The probabilities of the intensity values are listed in Table 1.

Intensity value	Count of occurrence	Probability of occurrence
20	2	1/32
21	6	3/32
22	6	3/32
54	2	1/32
55	2	1/32
93	2	1/32
94	2	1/32
95	8	1/8
165	2	1/32
166	8	1/8
167	4	1/16
253	6	3/32
254	8	1/8
255	6	3/32

Table 1

The entropy of the image is estimated as

$$\begin{aligned}
 H &= - \sum_{k=0}^{255} p_r(r_k) \log_2 p_r(r_k) \\
 &= - \left[6 * \frac{1}{32} \log_2 \frac{1}{32} + \frac{1}{16} \log_2 \frac{1}{16} + 4 * \frac{3}{32} \log_2 \frac{3}{32} + 3 * \frac{1}{8} \log_2 \frac{1}{8} \right] \\
 &= 3.5931 \text{ bits/pixel}
 \end{aligned}$$

- (b)

Figure 1 shows one possible Huffman source reduction and code assignment. Use the procedures described in the lecture and textbook. The intensities are first arranged in order of probability. The least probable symbols are then combined to create a reduced source and the process is repeated until the summation of probabilities equals to 1. Code words are then assigned.

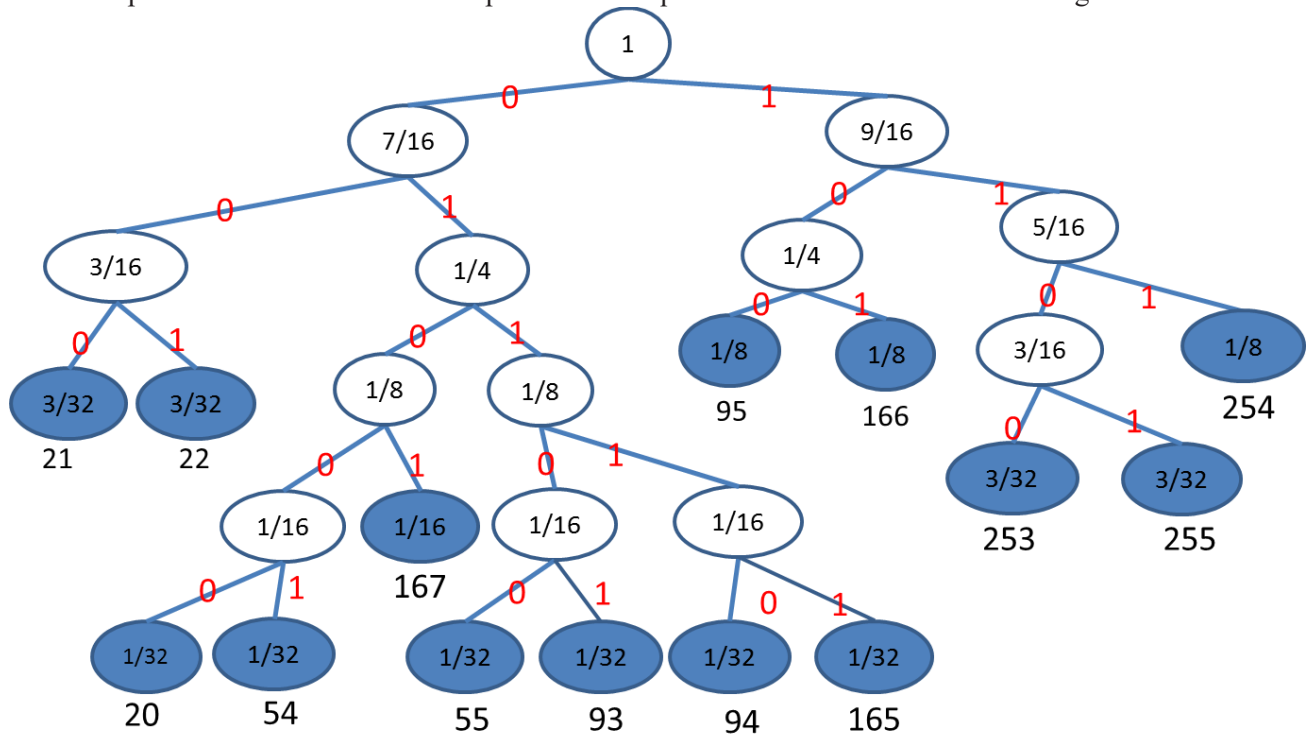


Figure1 Huffman Coding Diagram

Hence, the code words are listed in Table 2.

Intensity value	Probability of occurrence	Code word
20	1/32	01000
21	3/32	000
22	3/32	001
54	1/32	01001
55	1/32	01100
93	1/32	01101
94	1/32	01110
95	1/8	100
165	1/32	01111
166	1/8	101
167	1/16	0101
253	3/32	1100
254	1/8	111
255	3/32	1101

Table 2

$$L_{avg} = 6 * \frac{1}{32} * 5 + \frac{1}{16} * 4 + 2 * \frac{3}{32} * 3 + 2 * \frac{3}{32} * 4 + 3 * \frac{1}{8} * 3 = 3.625 \text{ bits/pixel}$$

The compression ratio is $8/3.625 = 2.207$ using Huffman coding.

- (c) We can have several solutions to problem using a combination of different compression methods. Obviously, the 8-bit image in the problem contains noise. Hence, we can first requantize the image as 6 digits and then apply Huffman coding on the requantized image.

Intensity value	Probability	New Intensity Binary	New Probability	New Code Word
20	1/32	000101	7/32	10
21	3/32			
22	3/32			
54	1/32	001101	1/16	000
55	1/32			
93	1/32	010111	3/16	001
94	1/32			
95	1/8			
165	1/32	101001	7/32	11
166	1/8			
167	1/16			
253	3/32	111111	5/16	01
254	1/8			
255	3/32			

Table 3

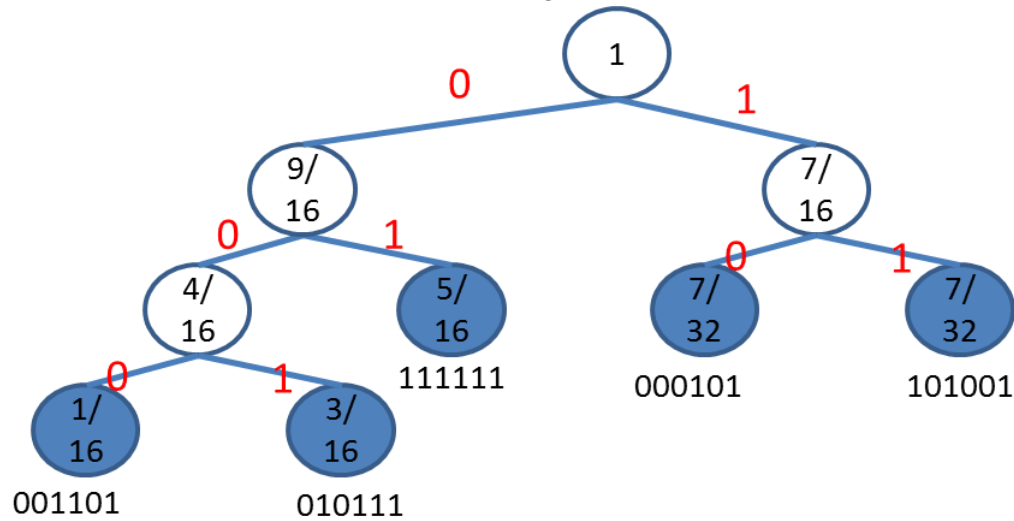


Figure2 Huffman coding diagram on the requantized image

$$L_{avg} = \frac{7}{32} \times 2 + \frac{1}{16} \times 3 + \frac{3}{16} \times 3 + \frac{7}{32} \times 2 + \frac{5}{16} \times 2 = 2.25 \text{ bits/pixel}$$

The compression ratio is $8/2.25 = 3.56$, which is better than that of the original Huffman coding.

2. The arithmetic decoding process is the reverse of the encoding procedure. Decode the message 0.222355 given the coding model. Assume that “!” is the symbol indicating the end of message. Show the decoding process. (25 pts)

symbol	probability
A	0.2
B	0.2

C	0.1
D	0.3
E	0.1
!	0.1

Solution:

First, the interval [0,1] is divided into 6 regions as shown in Table 5 based on the probabilities of each source symbol given in Table 4.

symbol	probability	Subintervals
A	0.2	[0, 0.2)
B	0.2	[0.2, 0.4)
C	0.1	[0.4, 0.5)
D	0.3	[0.5, 0.8)
E	0.1	[0.8, 0.9)
!	0.1	[0.9, 1.0]

Table 5

Step1. $V_1 = 0.222355$. $s_1 = B$ since $V_1 \in [0.2, 0.4)$

Step2. $V_2 = \frac{v_1 - \text{low}(s_1)}{p(s_1)} = \frac{0.222355 - 0.2}{0.2} = 0.111775$. $s_2 = A$ since $V_2 \in [0, 0.2)$

Step3. $V_3 = \frac{v_2 - \text{low}(s_2)}{p(s_2)} = \frac{0.111775 - 0}{0.2} = 0.558875$. $s_3 = D$, since $V_3 \in [0.5, 0.8)$

Step4. $V_4 = \frac{v_3 - \text{low}(s_3)}{p(s_3)} = \frac{0.558875 - 0.5}{0.3} = 0.19625$. $s_4 = A$, since $V_4 \in [0, 0.2)$

Step5. $V_5 = \frac{v_4 - \text{low}(s_4)}{p(s_4)} = \frac{0.19625 - 0}{0.2} = 0.98125$. $s_5 = !$, since $V_5 \in [0.9, 1.0]$

The message is "BADA!", if "!" is the symbol indicating the end of message.

3. A binary image contains straight lines oriented horizontally, vertically, at 45° , and at -45° . Give a set of 3x3 masks that can be used to detect 1-pixel breaks in these lines. For example, you may want to detect a pattern like 1 1 1 0 1 1 1 in the horizontal direction. Assume that the intensities of the lines and background are represented by 1 and 0, respectively. (25 pts)

Solution:

The masks would have the coefficients shown in Fig. 3. Each mask would yield a value of 0 when centered on a pixel of an unbroken 3-pixel segment oriented in the direction favored by that mask. Conversely, the response would be a +2 when a mask is centered on a one-pixel gap in a 3-pixel segment oriented in the direction favored by that mask.

0	0	0	0	1	0	0	0	1	1	0	0
1	-2	1	0	-2	0	0	-2	0	0	-2	0
0	0	0	0	1	0	1	0	0	0	0	1
Horizontal			vertical			45°			-45°		

Figure3. Masks to detect 1-pixel breaks

4. The results obtained by a single pass through an image of some 2D masks can be achieved also by two passes using 1D masks for improving efficiency. Show that the response of Sobel masks

$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$) can be implemented similarly by one pass of a vertical differencing mask $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ followed by a horizontal smoothing mask $[1 \ 2 \ 1]$. (25 pts)

Solution:

Suppose we have a subimage $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ under the Sobel mask $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$, where the mask is centered at the pixel with value “ e ”. The response is $-a - 2b - c + g + 2h + i$. If we perform two passes using 1D masks on the same subimage, we will first get $[g - a \quad 2(h - b) \quad i - c]$ with the mask $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, and then get $-a - 2b - c + g + 2h + i$ with the mask $[1 \ 2 \ 1]$. We can see that the result is exactly same with the one obtained by the 2D mask.

Same conclusion can be achieved for the mask $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$.